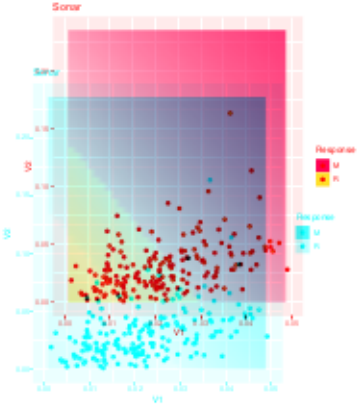
Introduction to Machine Learning

Classification: Tasks

Tasks



Learning goals

- Understand the main difference between regression and classification
- Know that classification can be binary or multiclass
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MULTICLASS TASK - IRIS

The iris dataset was introduced by the statistician Ronald Fisher and is one of the most frequent used data sets. Originally, it was designed for linear discriminant analysis.

Setosa

Versicolor

Virginica

Source: https://en.wikipedia.org/wiki/Iris_flower_data_set
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MULTICLASS TASK - IRIS

- 150 iris flowers
- Predict subspecies
- Based on sepal and petal length / width in [cm]

```
##     Sepal.Length Sepal.Width Petal.Length Petal.Width  Species
##  1:          5.1         3.5          1.4         0.2   setosa
##  2:          4.9         3.0          1.4         0.2   setosa
##  3:          4.7         3.2          1.3         0.2   setosa
##  4:          4.6         3.1          1.5         0.2   setosa
##  5:          5.0         3.6          1.4         0.2   setosa
## ---
## 146:          6.7         3.0          5.2         2.3 virginica
## 147:          6.3         2.5          5.0         1.9 virginica
## 148:          6.5         3.0          5.2         2.0 virginica
## 149:          6.2         3.4          5.4         2.3 virginica
## 150:          5.9         3.0          5.1         1.8 virginica
```